

Joseph Fajen

Documentation Systems Architect, Developer Documentation Specialist, Senior
Technical Writer

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- 20+ years of technical writing experience, focused on writing documentation, designing information architecture, and thinking about strategic content – transforming disparate communications into a single source of truth that's easy to search, navigate, and update.
- Deep experience designing and building documentation tools from the ground up, integrating AI and LLMs, generating content, analyzing output, and validating information accuracy.
- Experienced at engaging with a variety of audiences, from domain experts in AI and blockchain to high school students building their first application.

TECHNICAL SKILLS

Documentation: HTML, CSS, Markdown, Docusaurus, MDX, OpenAPI/Swagger

Development: Python, TypeScript, JavaScript, React, Next.js, Git/GitHub, GitHub Actions

AI: Claude Code, LLM APIs, Vector Databases, RAG Pipelines, Pydantic AI, Claude Agents SDK, MCP Servers

Domains: AI & LLMs, Blockchain & Smart Contracts, REST APIs, Video Systems

EXPERIENCE

Senior Technical Writer, Input Output Global

2022–2025

Input Output Global (IOG), originally IOHK, is the blockchain research and engineering company that created and maintains the Cardano protocol.

- Authored and curated 17,900+ lines of documentation for a smart contract platform across 124 files over 13 months, accounting for 78% of repository commits and 59 merged PRs
- Designed information architecture with 5 user personas and visual pathway diagrams; collaborated with engineers to integrate 91-schema OpenAPI specification and 41-page tutorial series
- Built full-stack AI assistant indexing 2,500 documents from 8 sources, achieving zero detectable hallucination through 20+ prompt engineering iterations

Senior Technical Writer, AJA Video Systems

2016–2022

AJA Video Systems designs and manufactures professional video hardware and software for broadcast, post-production, streaming, and ProAV markets.

- Produced 70+ product manuals (30–120 pages each) over 6 years, including 120-page flagship guides for Bridge Live and Bridge NDI 3G — the company's most complex multi-path streaming systems; reduced support ticket volume by enabling comprehensive customer self-service
- Proactively initiated the company's first REST API documentation covering 12 products with 500+ operations, partnering with 2 engineers through iterative reviews to deliver developer resources that previously didn't exist
- Validated documentation accuracy by configuring multi-product setups daily (encoders, routers, converters, monitors), tracking signal flows across 4+ devices; served as informal QA, catching errors before release

Senior Technical Writer, Ensemble Designs

2008–2016

Ensemble Designs manufactures video signal processing and routing equipment for broadcast, post-production, and live event production.

- Documented 80+ signal processing products across Avenue and BrightEye lines, evolving templates as complexity grew; engaged directly with customers at NAB tradeshow to validate documentation approach, receiving positive feedback
- Sole writer for 6 flagship router products featuring industry-first configurable I/O — a new paradigm replacing fixed-matrix architectures; created manuals from scratch that enabled sales and drove customer adoption

OTHER PROJECTS

ISEE v2 — Idea Synthesis and Extraction Engine (2026)

Built a multi-model AI orchestration system that expands the possibility space through combinatorial synthesis (6 LLMs × 11 cognitive frameworks), then extracts breakthrough ideas through structured tournament debate. Includes operations dashboard with real-time cost tracking and observability. Project at: github.com/joseph-fajen/isee-v2.

Book of Archon (2026)

Created a 10-chapter official onboarding guide for Archon (archon.diy), a harness-building system for AI coding agents. Built using a custom adversarial AI workflow—Writer Agent vs. Attacker Agent with 7/10+ quality gates—ensuring rigorous accuracy and clarity. Submitted via PR, accepted and merged, now serving as the project's official user guide. Archon has 20k+ stars on GitHub.

Programming Tutor, CodeHS (2013–2015)

CodeHS is an online platform teaching computer science fundamentals to students through interactive JavaScript courses. Evaluated course materials, reviewing and completing 50+ coding exercises for JavaScript proficiency.

EDUCATION

Agentic Coding Course — Dynamous, 2025–2026

Systematic methodology for AI-assisted development: structured prompt templates, deliberate planning, and context engineering using the PIV Loop (Plan → Implement → Validate) framework.

Hands-On Linux System Administration — Packt, 2024

Hands-on training in server administration, package management, Apache configuration, and system monitoring—foundational skills for managing documentation infrastructure and understanding production environments.

The Git & Github Bootcamp — Udemy, 2023

Comprehensive coverage of Git workflows including branching, merging, rebasing, and GitHub collaboration patterns essential for docs-as-code infrastructure.

B.A., History and Philosophy — U.C. Davis